4.4 The black body

4.4.1 *Definition of a black body*

If different stars have different colors, the colors must tell us something about the nature of the stars and their differences. In order to find out about the possible meaning, we have to compare the colors of the stars with colors of known objects on Earth. We do not learn anything if we compare the stars with just *any* colored object on Earth, we need some sort of *ideal source of light* whose properties depend only on one parameter. Such a source is the so-called 'black body', whose radiation depends only on the temperature.

What is a black body? We would call an object black if it looks black in bright daylight, this means that the object does not send out any light rays that enter our eyes. The light that our eyes receive from objects which do not look black in bright daylight is usually not light which these objects emit of their own but is solar light that is reflected by these objects. Solar light appears yellow to us. Most objects on Earth do not look yellow even though we see only solar light reflected by them. The reason for the difference in color is the fact that any object absorbs part of the solar light, and only the part which is not absorbed can be reflected. If mainly blue wavelengths were absorbed, then the object will appear more red than the sun, if mainly red wavelengths were absorbed then the object will appear more blue. If all wavelengths are strongly absorbed, then no light is left to be reflected into our eyes and the object will appear black. We therefore define a *black body* as *an object that absorbs all light* which falls on it. This does not mean that a black body always has to look black. A black body may generate radiation by itself, then it may actually be quite bright even though it absorbs all the light reaching it from other light sources. Think about a hot plate on an electric stove. If it is not turned on, it looks almost black because it absorbs nearly all the light which falls on it, but when you turn it on, it becomes hot and now produces its own light. It then shines brightly even if you turn off all the outside lights. The hot plate is still nearly a black body because it still absorbs nearly all the light falling on it but it certainly does not look black any more. When you turn the hot plate on 'LOW' you will see it glow in a dark red color, when you turn it on higher you will find that it becomes brighter and more yellowish; if it becomes still hotter, it will shine still brighter and more bluish. In this example, we see already that the color of such a near black body tells us something about the temperature of the object.

How can we realize an ideal black body, which means an object that

absorbs all the light falling on it? The best way to do this is to make a small hole in a box, see Fig. 4.6. A light beam falling into the hole will hit a wall in the box and will be reflected, part of it may be absorbed by the wall. The reflected part will hit another wall and will be partly absorbed and partly reflected again and so on. Chances that even a small fraction of the light might finally find its way back out through the hole are extremely small and will be smaller the smaller the hole is. The hole is indeed a black body. Actually the windows in a building are approximations to this kind of a black body; when you look at the windows from the outside, they appear black, except if you put a light source inside this black body, such as a burning electric bulb, then the black body radiates on its own merits.

If we insulate such a black body very well from its surroundings and leave it to itself for a long time, then the inside of the box, shown in Fig. 4.6, achieves an equilibrium, which means that nothing in the box will change anymore in time or in space. We say the box has reached thermodynamic equilibrium. In physics we only call a box with a hole a black body if it has achieved thermodynamic equilibrium. This is included in the definition of a 'black body.'

4.4.2 *Radiation of a black body*

For a black body we can measure the radiation that comes out of the hole, it is radiation emitted by the black body itself, either by gas in it or by the inside walls of the box. For the radiation of such a black body in thermodynamic equilibrium we can measure that the radiation coming out of the hole is independent of the material of the box and independent of the gas in the box. No matter which kind of box we take, *after thermodynamic equilibrium is achieved* the radiation is always the same if the temperature of the box is the same, but the radiation changes if the temperature of the black

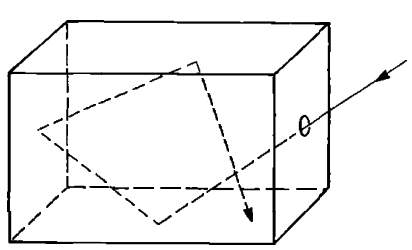

Fig. 4.6. A black body can best be realized by a small hole in a box. Light falling into the hole in the box will be reflected back and forth and will not find its way out of the hole before being absorbed at the walls. For a black body, as defined in physics, we have to insulate the box very well from its surroundings and leave it to itself for a long time so that thermodynamic equilibrium can be achieved in the box.

body changes. The change is similar to the changes seen for the hot plate with increasing temperature. For higher temperatures, more radiation is emitted and the radiation looks more blue. From such measurements of the radiation of a black body we find that the total radiative energy E emitted per cm^2 and per s in all directions by a black body of temperature T increases with the fourth power of the temperature, i.e.

$$E = \sigma T^4 = \pi F. \tag{4.14}$$

where σ is the so-called Stefan–Boltzmann constant and is $\sigma = 5.67 \times 10^{-5}\,\mathrm{erg\,cm^{-2}\,s^{-1}\,K^{-4}}$, F is called the flux. Equation (4.14) is the *Stefan–Boltzmann law.*

4.4.3 *Spectral energy distribution*

If we measure the amount of radiation per wavelength interval of 1 cm coming out of the hole per s perpendicular to the area of the hole and into a cone with a solid angle $\Delta\omega = 1$, and if we divide this amount by the area of the hole, then this energy is called the intensity I_λ (see Fig. 4.7)

In Fig. 4.8 we show the wavelength dependence of the intensities emitted by black bodies of different temperatures.

Of course, we cannot make black bodies with temperatures of 6000 K in the laboratory because any box of any kind of material would melt at such temperatures, but we can derive the law for the intensity distribution from measurements for lower temperatures and then use this law to calculate what the energy distribution for higher temperatures will look like. You will recall that it was the observed law for the radiation of a black body which led Max Planck to the discovery of the light quanta.

For a black body the intensity $I_\lambda = B_\lambda$ is given by

$$I_\lambda = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT} - 1} = B_\lambda = \text{Planck function.} \tag{4.15}$$

Here, c = velocity of light, k = Boltzmann constant, and h = Planck constant.

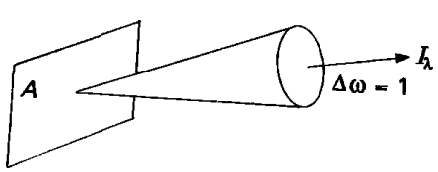

Fig. 4.7. The energy emitted from a surface area A of 1 cm^2 per s per wavelength band $\Delta\lambda = 1$, perpendicular to A and into a cone with opening $\Delta\omega = 1$, is called the intensity I_λ. For a black body the intensity $I_\lambda = B_\lambda =$ Planck function, given by (4.15).

We certainly do not measure the intensity with a wavelength band of 1 cm because the intensity would vary by very large factors over such a wide band. What we mean is that we measure the intensity in a small wavelength band $\Delta\lambda$ and then divide the measured energy by the wavelength interval $\Delta\lambda$. The amount of energy received will of course increase if the bandwidth is increased.

Keeping in mind the energy distributions of black bodies and the increasingly more blue color for higher temperatures we may suspect that the blue stars have higher temperatures than the red stars.

4.4.4 *Wien's displacement law*

The intensity distribution of a black body as given by (4.15) has a very important property. As we see in Fig. 4.8, the maximum shifts to shorter wavelengths for higher temperatures. From (4.15) it can be easily derived that the wavelength λ_{max}, for which the maximum intensity is measured. (i.e. the wavelength λ_{max} for which $dI_\lambda/d\lambda = 0$) is inversely proportional to the temperature T, or in other words,

$$\lambda_{max} \cdot T = \text{const.} = 0.2897\,\text{cm}\cdot\text{degree} \tag{4.16}$$

if the wavelength λ_{max} is measured in cm.

Equation (4.16) offers an easy way to determine the temperature of a black body without having to measure the whole energy distribution, we only have

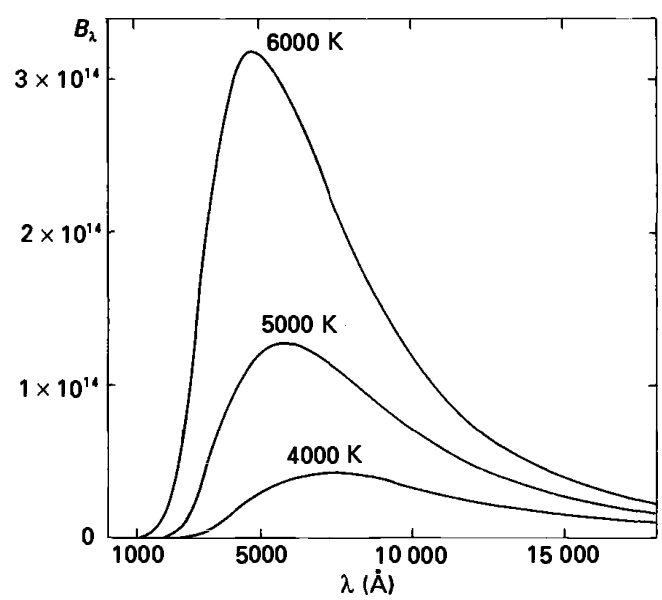

Fig. 4.8. The intensity $I_\lambda = B_\lambda$ is shown as a function of the wavelength λ for black bodies with different temperatures T, as indicated on the curves.

to determine the wavelength at which the energy distribution has its maximum. If the stars were to radiate like black bodies we could easily determine their temperatures in this way.

4.4.5 *The colors of black bodies*

Because black bodies become more blue with increasing temperature we suspect that the colors of the stars tell us something about the temperatures of the stars. We expect the blue stars to have higher temperatures than the red stars because for black bodies the radiative energy shifts to shorter wavelengths for higher temperatures. In order to check how well the colors of the stars agree with those of black bodies we have to determine the B – V and the U – B colors for black bodies with different temperatures. This means we have to determine the U, B, V magnitudes of the black bodies which in turn requires us to compare the brightness of the black bodies in the U, B and V bands with the brightness of Vega. Unfortunately, this is not very easy. In order to have the same transmission functions for the star and the black body we have to observe the star and the black body with the same instrument, namely with the telescope. The telescope gives only sharp images of light sources which are very far away. We therefore have to put the black body at a large distance which makes it rather faint.

Even more of a problem is the correction for extinction in the Earth's atmosphere. The beam of light from Vega passes through the whole atmosphere, all layers of the atmosphere with their different absorption coefficients contribute to the extinction. For the black body, which we may put at the next mountain top, only the lower layers of the atmosphere contribute to the extinction. We would have to know the height dependence of the atmospheric absorption coefficient in order to make the correct extinction correction for the black body. While from the physical point of view it would be more logical to compare the stellar brightnesses directly with those of black bodies to begin with, this causes so many difficulties that it is intrinsically much more accurate to compare stars with stars. The final comparison with black bodies does not then influence the intrinsic comparison of stars among themselves.

When we do the best we can for the determination of the colors of black bodies, we find the B – V and U – B colors for black bodies as shown in the two color diagram, Fig. 4.9. Table 4.2 shows the actual numbers. Every point on the black body curve in Fig. 4.9 corresponds to a black body with a given temperature, as indicated for a few values of the temperature. Remember

stars with positive values of B – V are more red than Vega. B – V therefore increases for lower temperature. The same holds for U – B.

In the same diagram we have also entered values measured for nearby stars. They also seem to follow one curve, which is, however, different from the one for black bodies, but still all stars with a given color B – V have nearly the same U – B color. Both colors are determined by one parameter, which we still suspect to be the temperature. Before we can, however, establish a relation between the colors of a star and its temperature we have to understand the difference between the stellar energy distributions and those of black bodies, and especially why the stars have so much less energy in the U band than the black bodies. This we will discuss in volume 2, when we will discuss the radiation and the structure of stellar atmospheres.

4.5 The solar radiation

The brightest star is of course our sun. We can measure the amount of radiative energy received from the sun and correct it for the absorption in

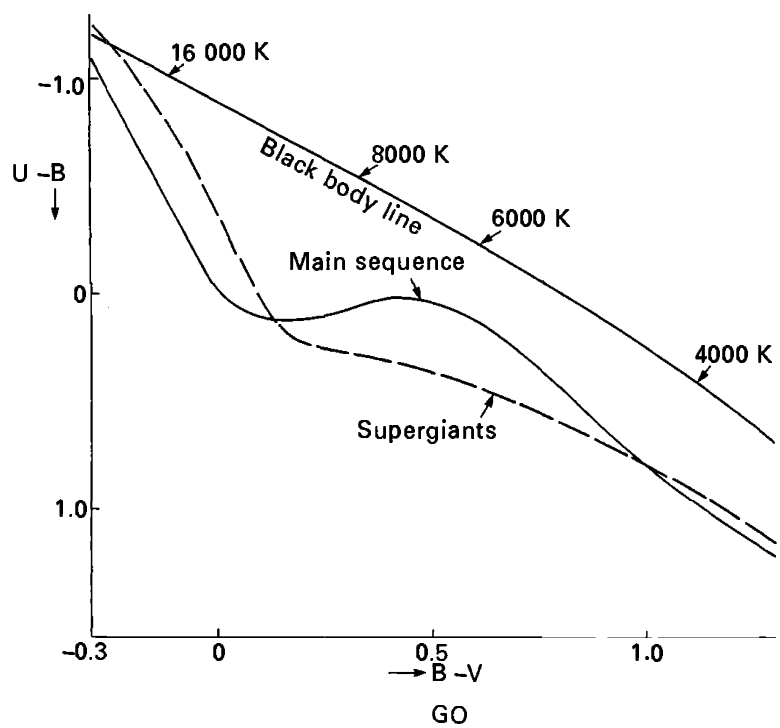

Fig. 4.9. The U, B, V colors for 'normal', main sequence stars are shown in a two-color diagram. For increasing B – V colors the U – B colors generally also increase, except in the range $0.1 < B - V < 0.5$. Also shown are the colors for supergaints, see chapter 10.2, for which the U – B colors always increase with increasing B – V. We have also plotted the U – B versus B – V colors for black bodies. For a given B – V the stars have less radiation in the ultraviolet than black bodies.

Table 4.2 *The B − V and U − B colors of black bodies with different temperatures according to Lamla, 1982*

B − V	U − B (± 0.03) Black body
−0.30	−1.22
−0.20	−1.11
−0.10	−1.00
0.00	−0.88
+0.10	−0.76
+0.20	−0.66
+0.30	−0.54
+0.40	−0.44
+0.50	−0.33
+0.60	−0.22
+0.70	−0.10
+0.80	+0.02
+0.90	+0.14
+1.00	+0.25
+1.10	+0.37
+1.20	+0.49
+1.30	+0.61

the Earth's atmosphere in exactly the way described in the previous section. We can now also measure the total solar radiation directly, outside the Earth's atmosphere from satellites. The satellites called 'Space Lab' and the 'Solar Maximum Mission' (the first satellite to be repaired while in orbit) have made very careful measurements of the solar radiation. What is found is an energy distribution, also shown in Fig. 4.5. The continuum energy distribution, which means the energy distribution between the spectral lines, looks very similar to that of a black body at 6000 K. The energy has a maximum around 5000 Å, the wavelengths at which our eyes are most sensitive, luckily enough also in the wavelength region in which the Earth's atmosphere is rather transparent.

How much energy do we actually get from the sun altogether? The total radiation received from the sun per s per cm^2, perpendicular to the beam of light above the Earth's atmosphere, is called the solar constant S. It is measured to be $S = 1.368 \times 10^6 \, erg \, cm^{-2} s^{-1}$.

Since there is so much talk about solar energy as the energy source for our

industrialized life, let us stop for a moment and realize how much that actually is.

Since 1 square foot has about $10^3\,\mathrm{cm}^2$ and since $10^7\,\mathrm{erg\,sec}^{-1}$ is 1 watt we can also write $S \approx 0.14 \times 10^3$ watt per ft^2 or $S \approx 0.14$ kW per ft^2.

If your room has a roof area of about $200\,\mathrm{ft}^2$ and the sun shines on average 5 hours per day, you would collect on your roof energy of the order of 140 kW hours each day. This is not really correct since the sun is not shining perpendicularly on your roof and there is some absorption in the atmosphere, but it gives the correct order of magnitude, it may be a factor of 2 or 3 less. This would still be plenty to cover your personal needs. Certainly for a large industrial plant the sun shining on their roof would not be sufficient, but our main problem still is that we do not yet know how to convert the solar ergs per s into kW of electrical energy efficiently. Currently we lose a factor of between 10 and 100 in the conversion.

4.6 The absolute magnitudes of stars

The apparent magnitudes do not really tell us anything about the intrinsic brightness of the stars which to the astrophysicist is much more important than the apparent brightness. A star may be intrinsically very bright and just appears very faint because it is so distant. If we can determine the distance we can correct for this effect. We can then in our minds put all the stars at the same distance and compare their intrinsic brightnesses. The magnitudes which the stars would have if they were at a distance of *10 pc* are called the *absolute magnitudes*. They are designated by a capital M. Again we have absolute magnitudes M_V, M_B, M_U for the different wavelength bands.

Vega is at a distance of 8.4 pc, i.e., is closer than 10 pc. If Vega was at a distance of 10 pc it would be fainter. Its absolute magnitude is therefore larger than its apparent magnitude: $M_V(\mathrm{Vega}) > 0$. It comes out to be $M_V(\mathrm{Vega}) = 0.5$. We have to calculate the magnitude which Vega would have if it were at the distance of 10 pc and compare this with the magnitude which it actually has, namely $m_v = 0$.

In order to determine the absolute magnitude we have to know the distance d of the star

$$\frac{I(10\,\mathrm{pc})}{I(d)} = \left(\frac{d}{10\,\mathrm{pc}}\right)^2 \quad \text{because} \quad I \propto \frac{1}{d^2} \tag{4.17}$$

$$M_V = m_v(10\,\mathrm{pc})$$

$$M_V - m_v = -2.5[\log I(10\,\mathrm{pc}) - \log I(d)]$$

$$= -2.5[2\log d - 2\log 10\,\mathrm{pc}]$$

$$= -5[\log d[\mathrm{pc}] - 1]$$

$$= -5\log d[\mathrm{pc}] + 5 \tag{4.18}$$

$m_v - M_V$ is called the distance modulus of the star because it is determined by the distance of the star.

For stars with parallax angles $\pi > 0''.05$ we can measure π fairly accurately. We can therefore determine their distances from their parallaxes. For these stars we can also determine their absolute magnitudes M_U, M_B, M_V.

What are the stellar colors in absolute magnitudes?

From (4.18) we see

$$M_V = m_v - 5\cdot\log d[\mathrm{pc}] + 5 \tag{4.19}$$

$$M_B = m_B - 5\cdot\log d[\mathrm{pc}] + 5 \tag{4.20}$$

and

$$M_B - M_V = B - V = m_B - m_V. \tag{4.21}$$

This means we can always determine the colors from the apparent magnitudes. The absolute colors are the same. See, however, Chapter 19 for interstellar reddening.